Lecture 1:Introduction

CS211 - Operating systems

Spring 2026

What do you know about Operating Systems & CS211?

1. When did you first hear the term “operating system”?
2. What is an OS?
3. Why do we need OSes?
4. What was the first OS?
5. How many OSes are being used actively?
6. Which OSes do you know or have used?
7. Which OS is the most popular now?
8. Which OS is the most commonly used on servers?
9. What is your favorite OS?
10. Is Windows the favorite OS of the CS211 instructor?
11. What makes a good OS?
12. Are people still developing new OSes?
13. Which programming languages are people using to develop new OSes?
14. How many lines of code are needed to write an OS?
15. Why do you take CS211?
16. What do you expect to achieve after completing CS211?
17. Which OS concepts did you know before enrolling in CS211?

What is an operating system?

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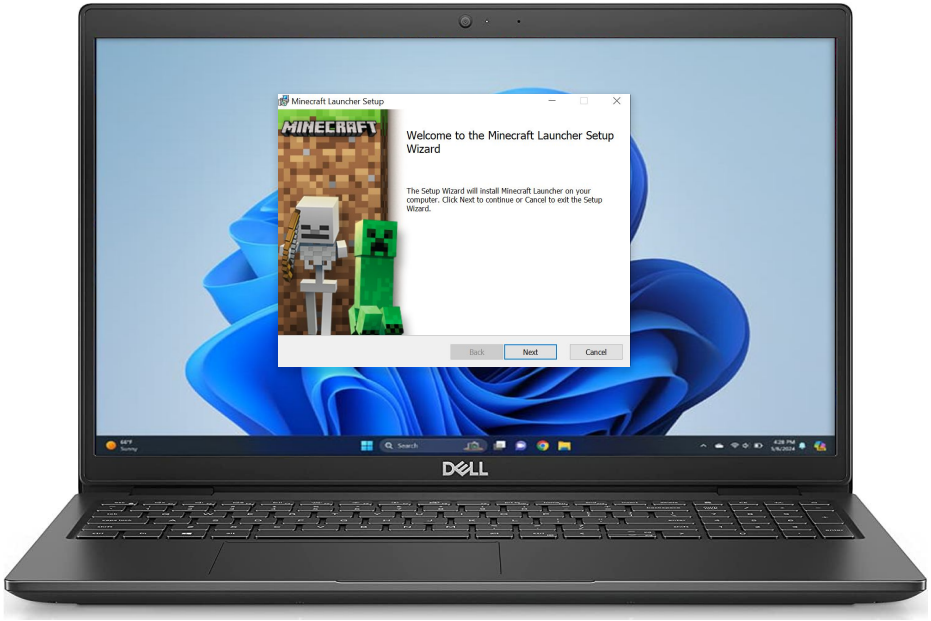

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	Hardware	Operating System	Application
Windows 11			
Minecraft			
Dell laptop			

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An operating system (OS) is system software that manages computer hardware and software resources, and provides common services for computer programs

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Hardware interface

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OS interface

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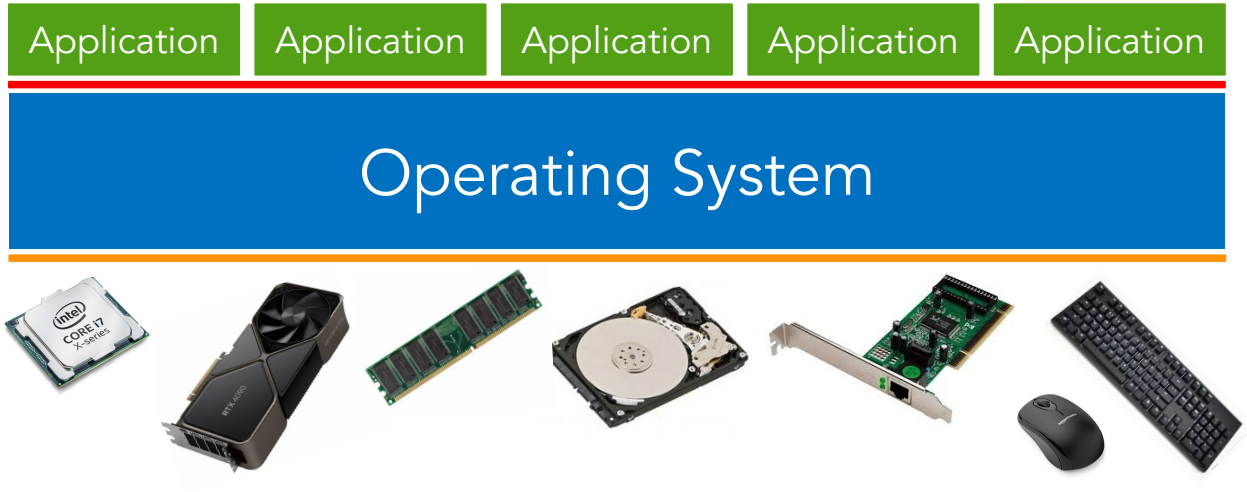

OS interface

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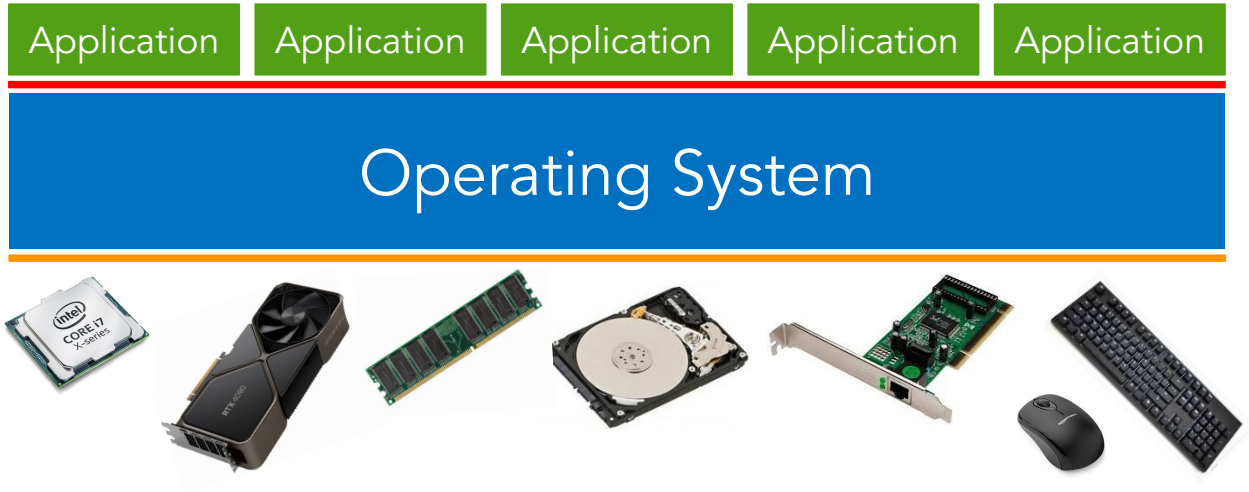

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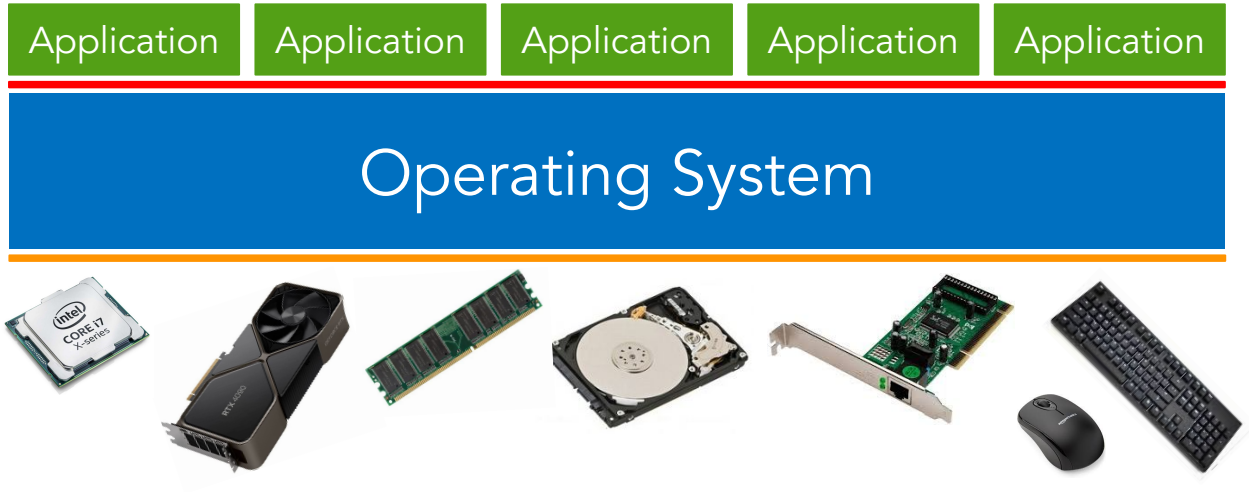

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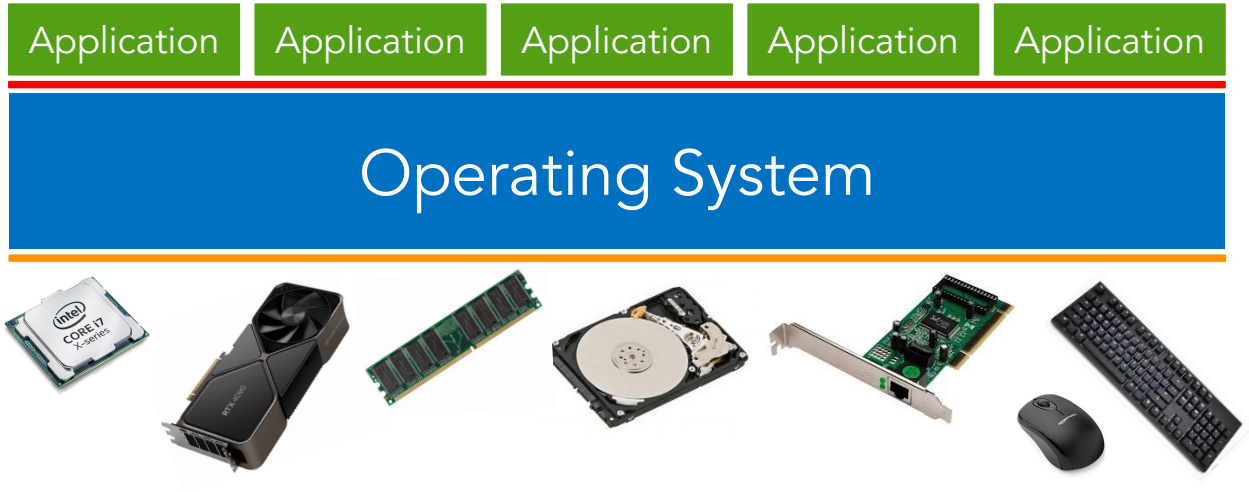

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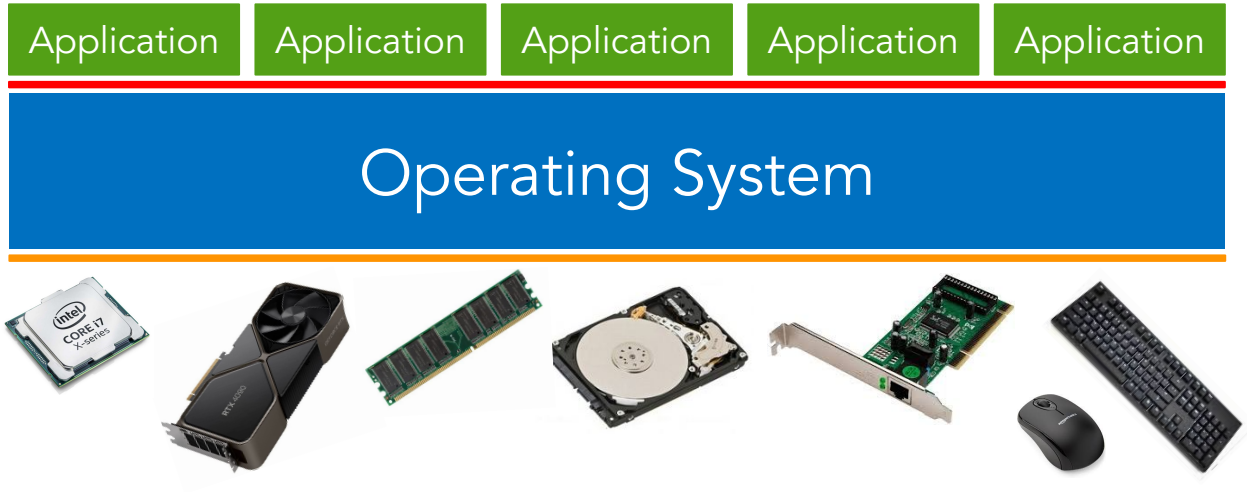

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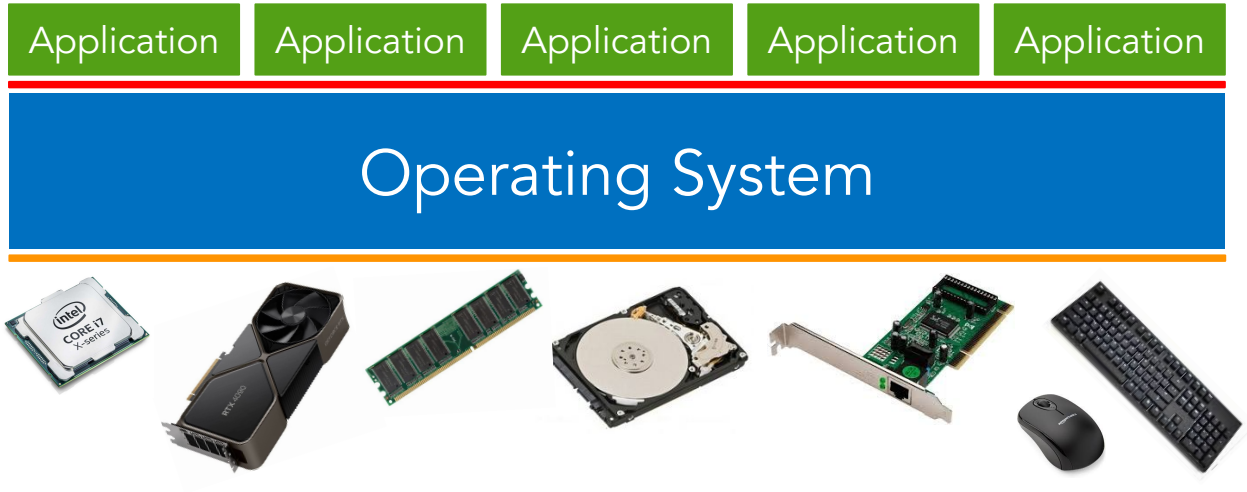

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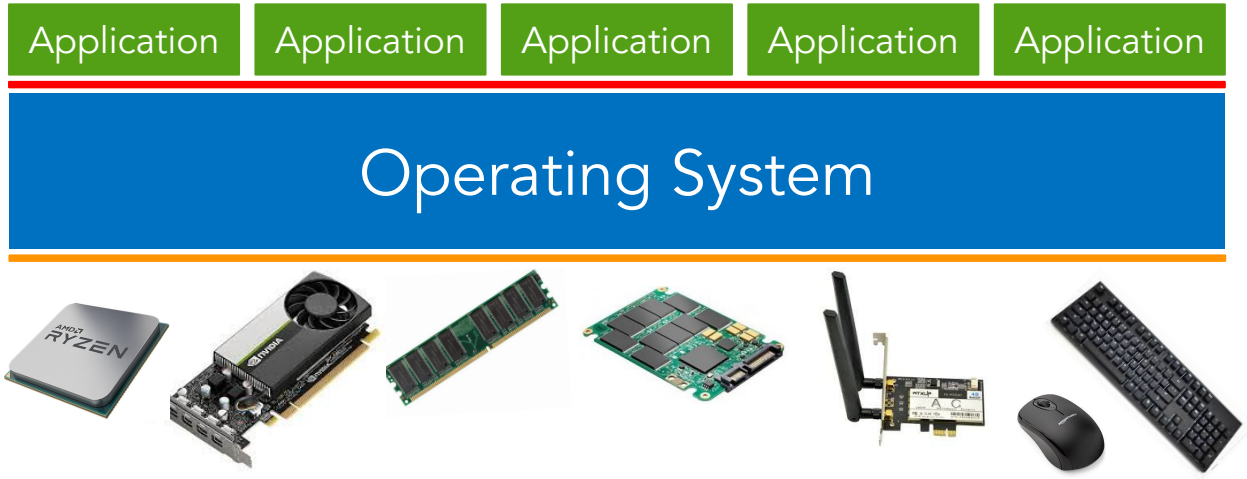

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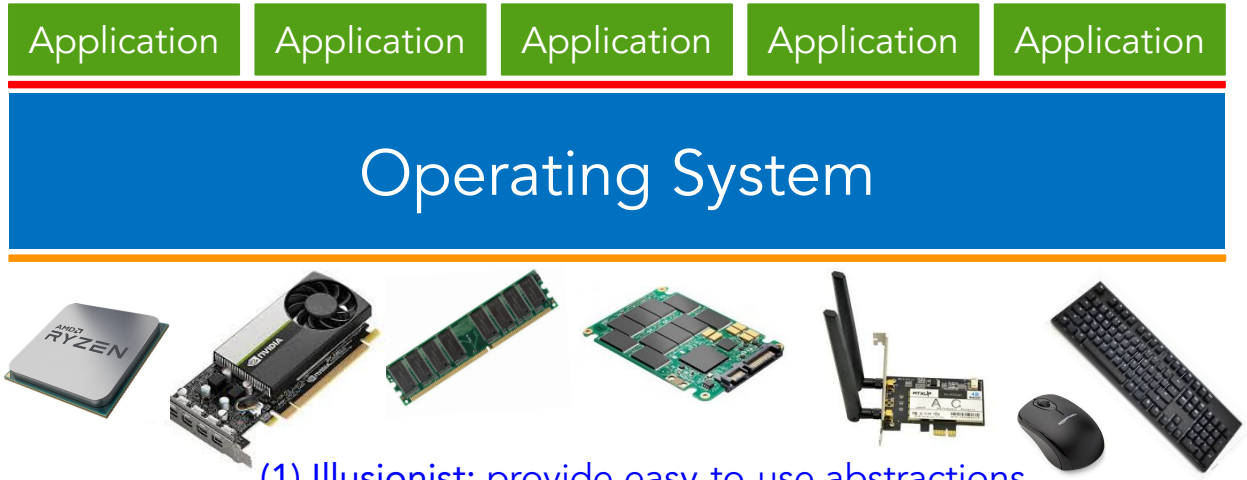

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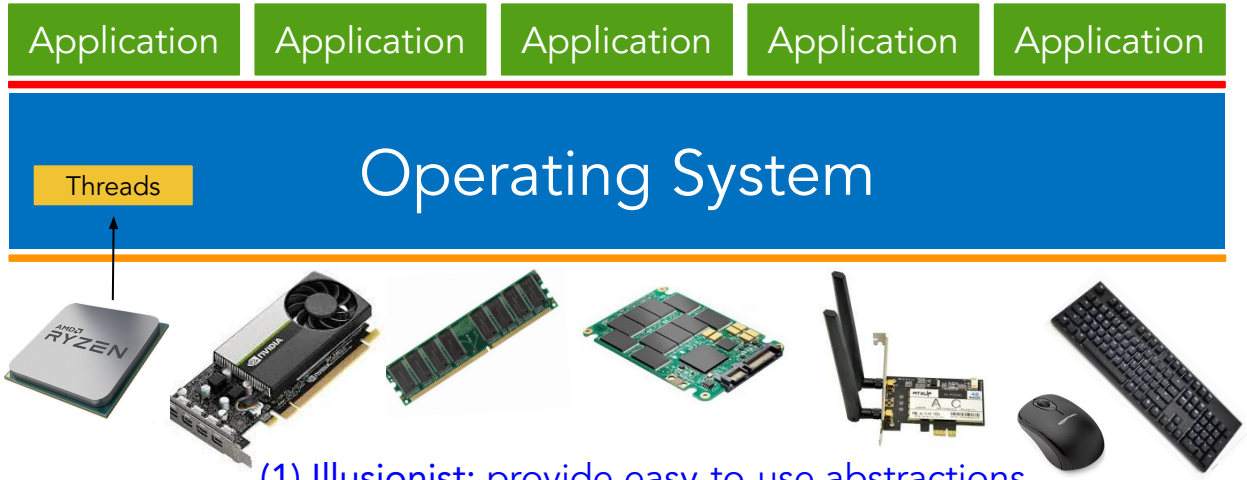

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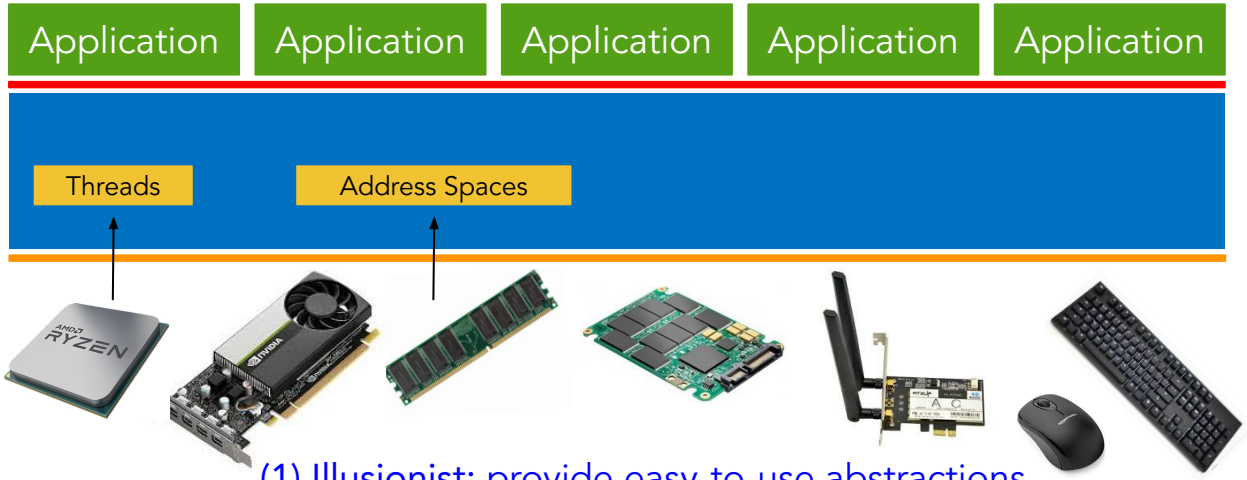

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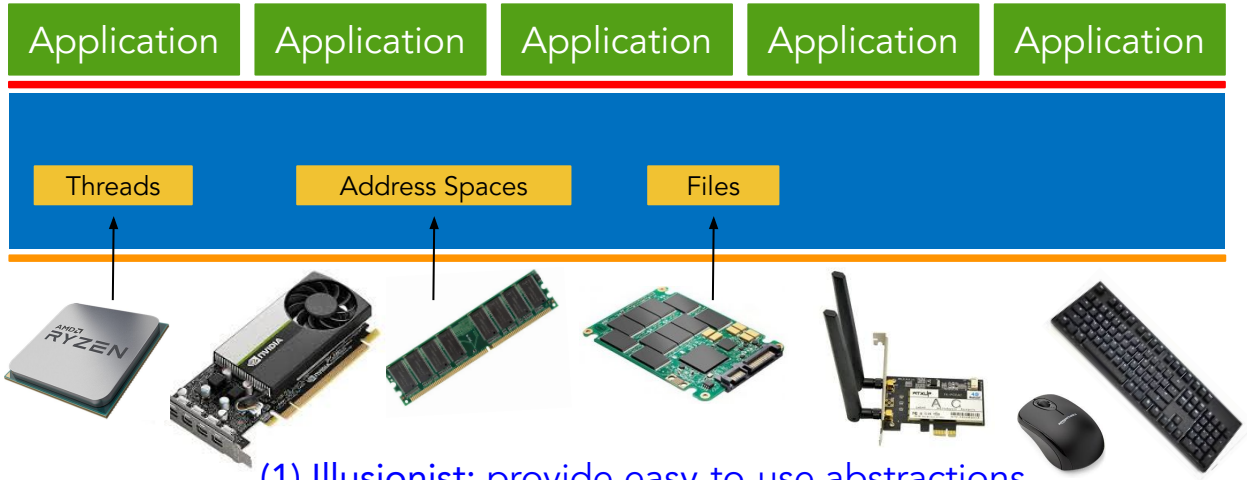

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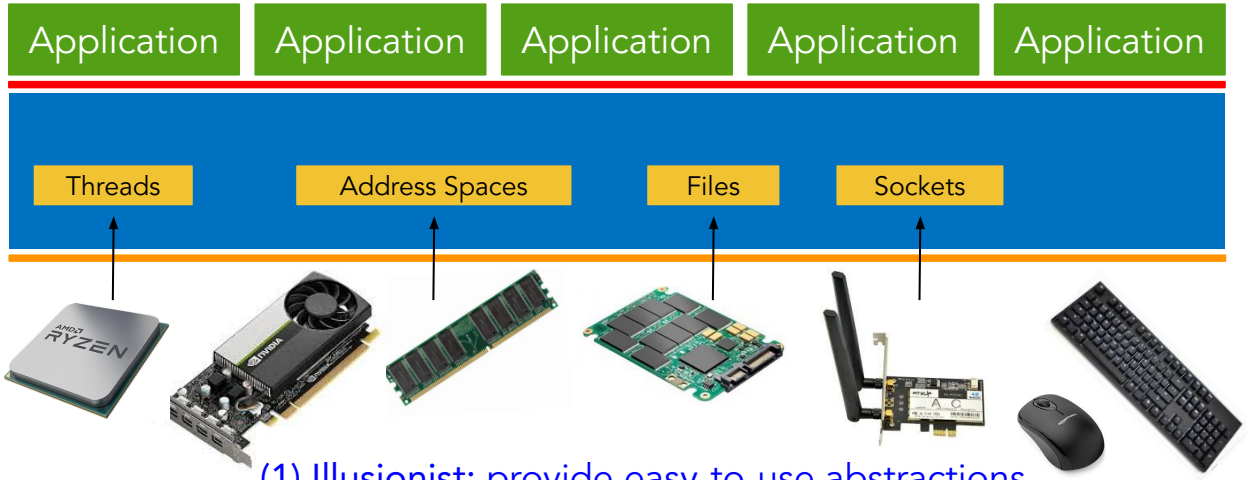

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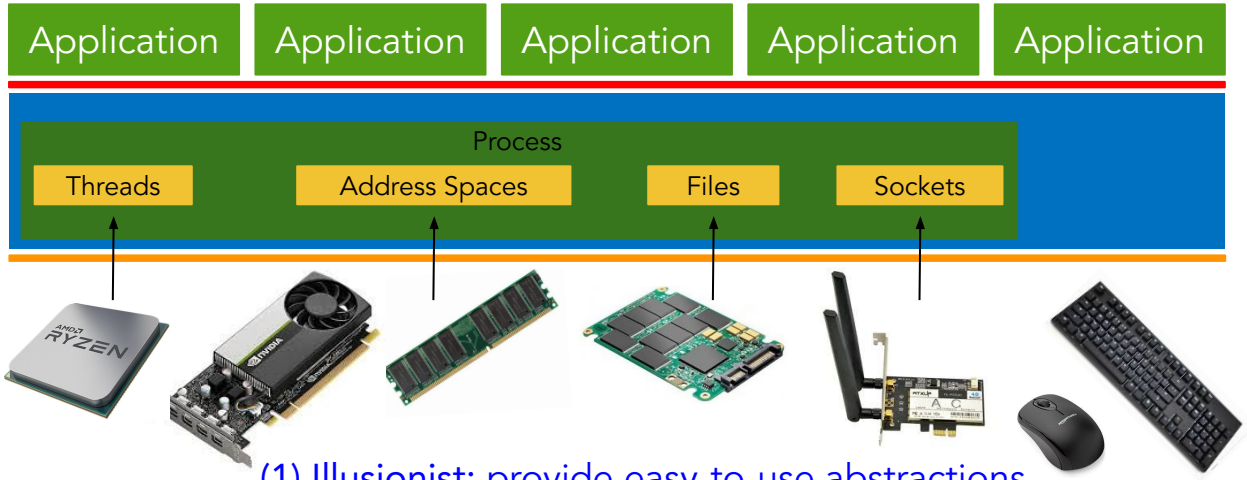

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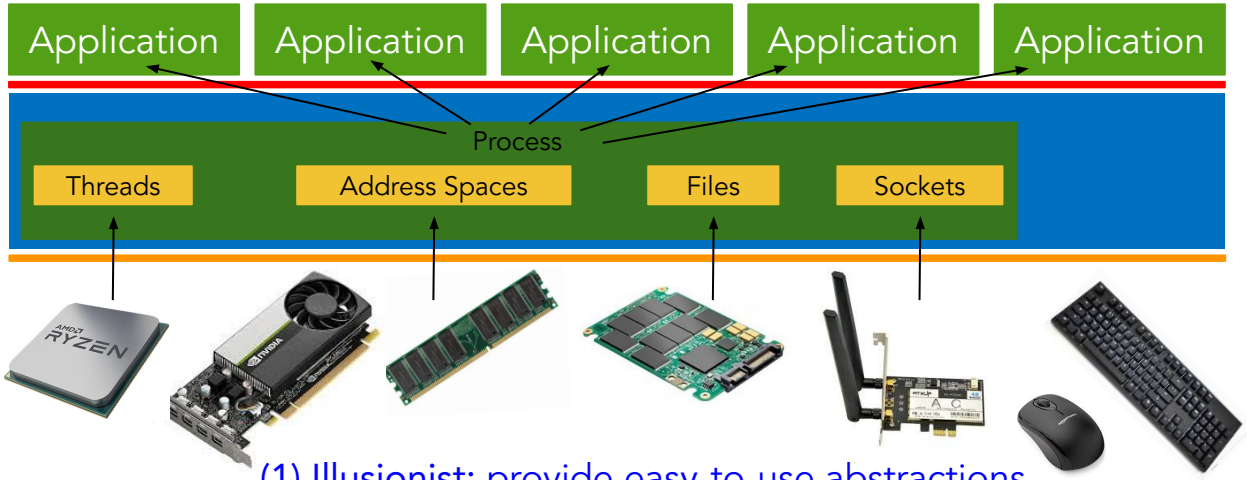

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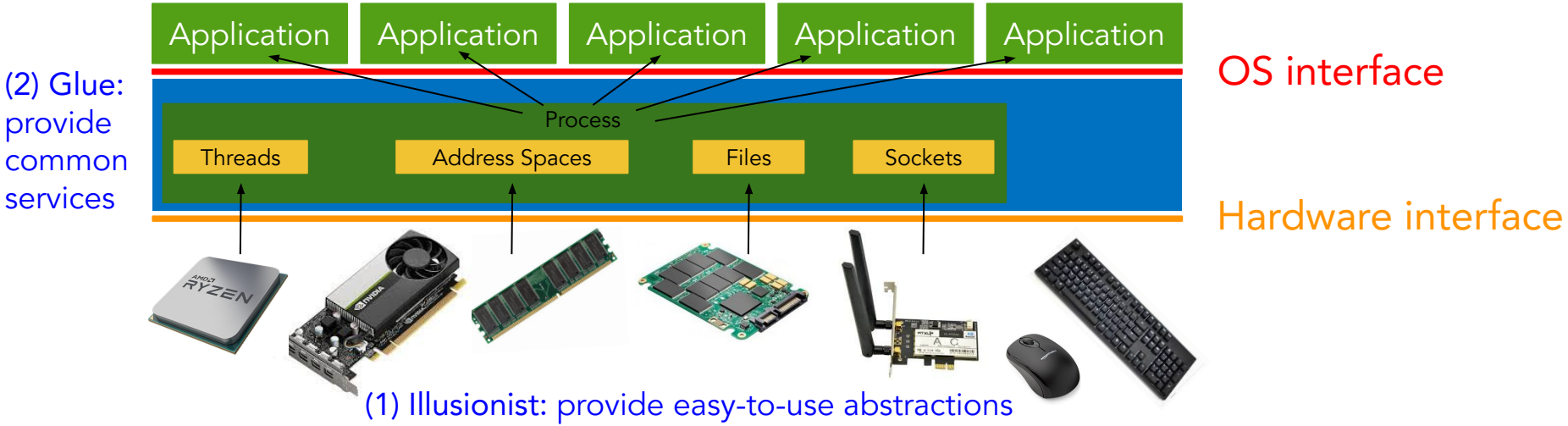

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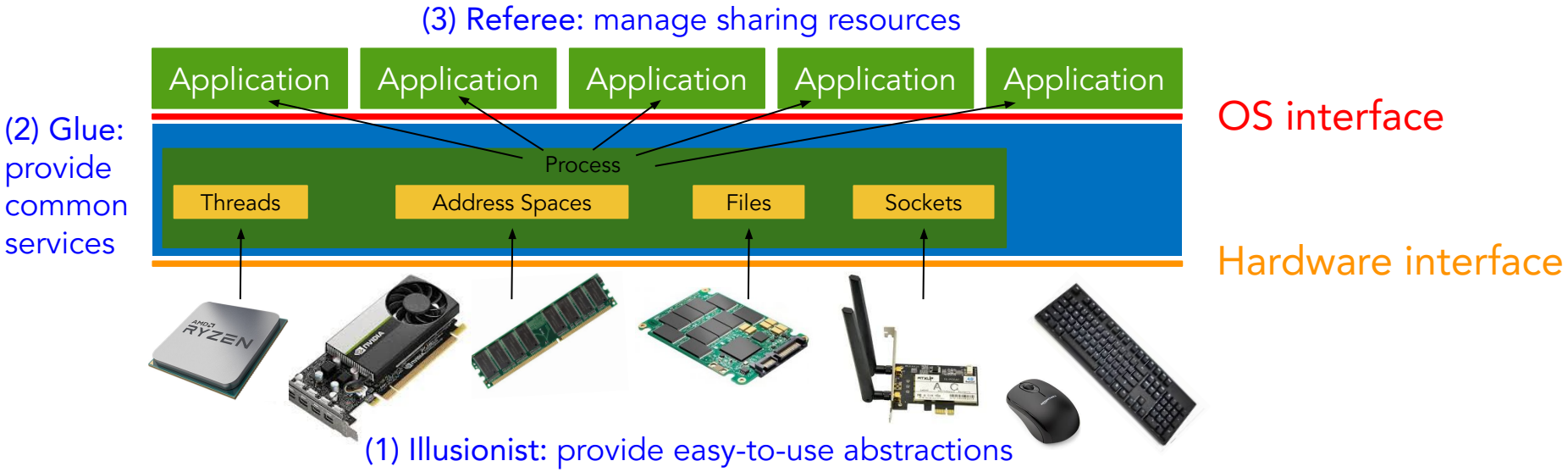

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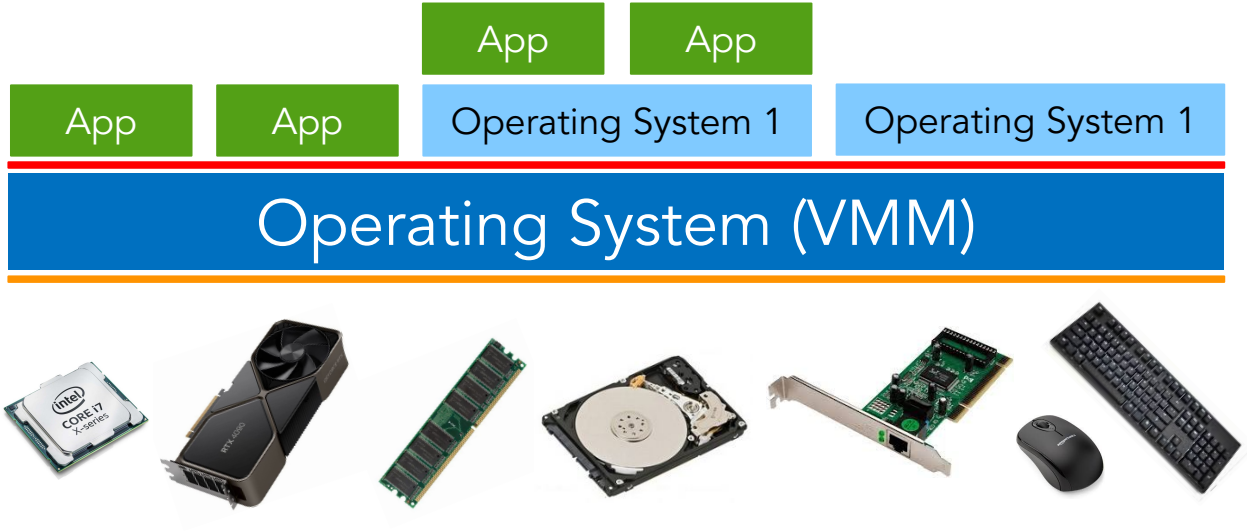

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Smart phones become popular after the iPhone was released in 2007

2020s

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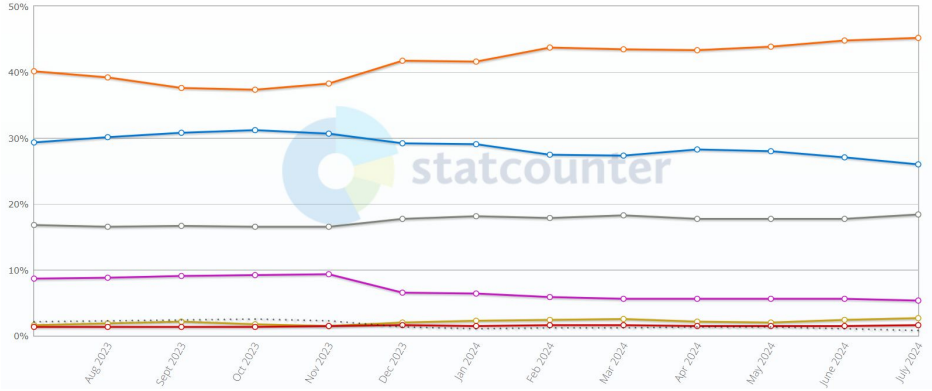

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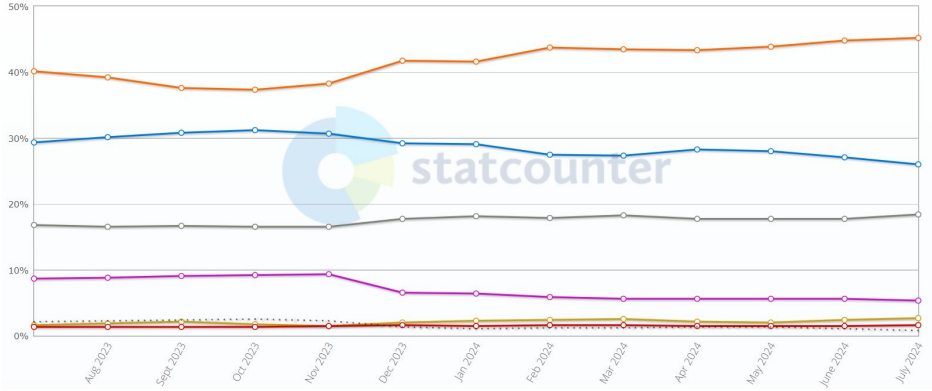

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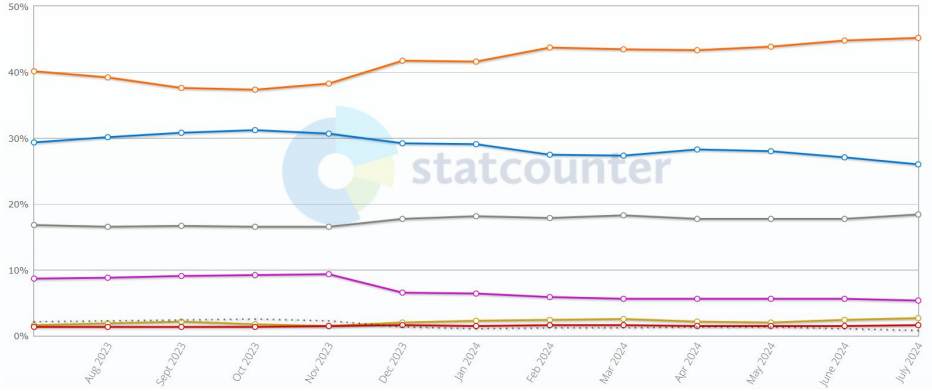

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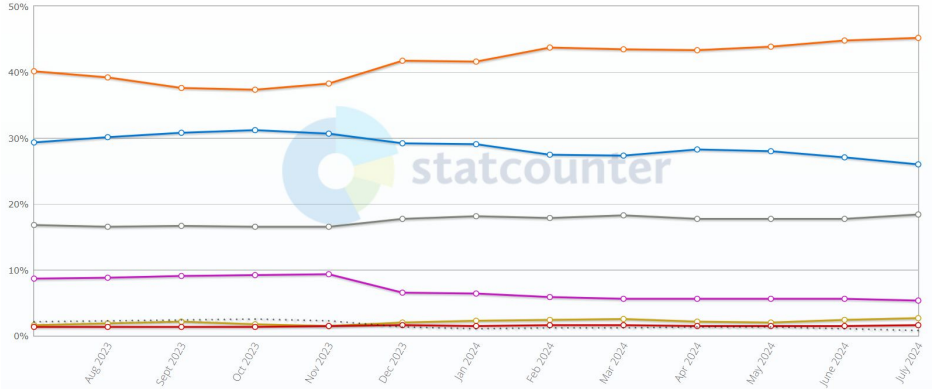

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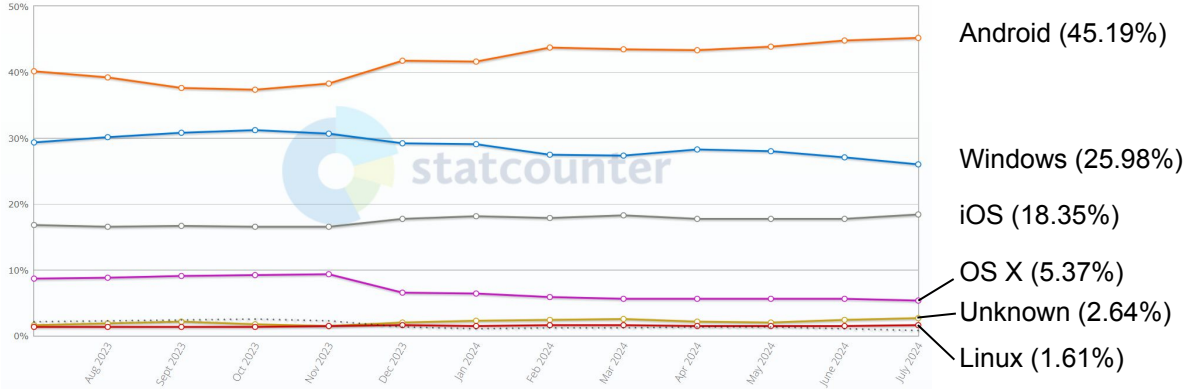

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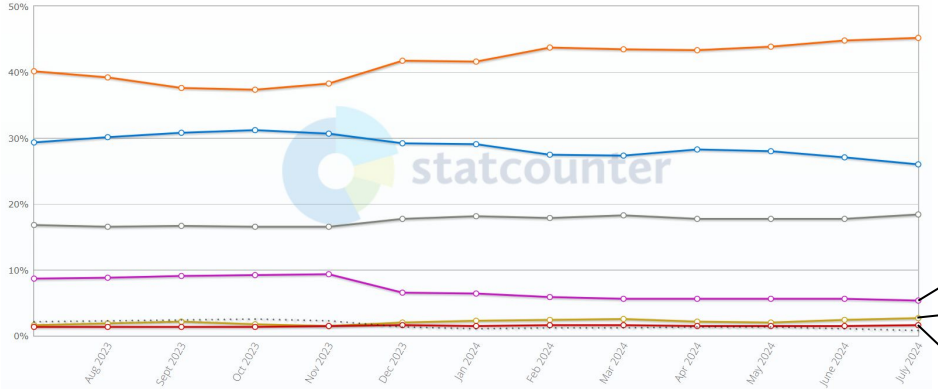

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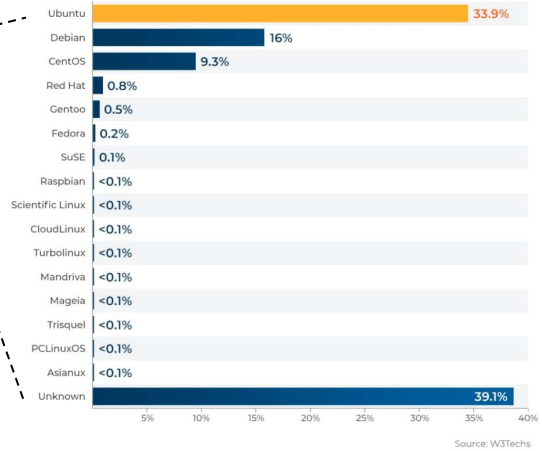
OS X (5.37%)

Unknown (2.64%)

Linux (1.61%)



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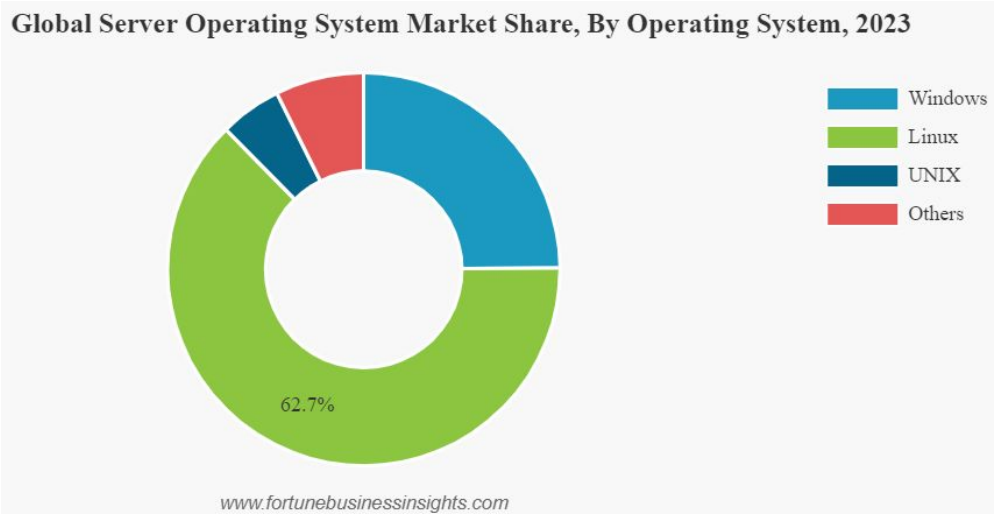

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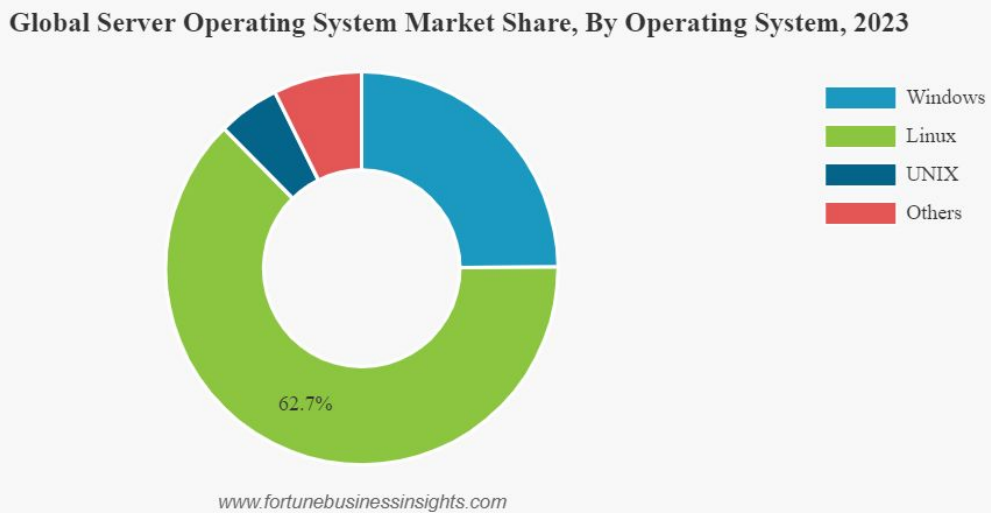

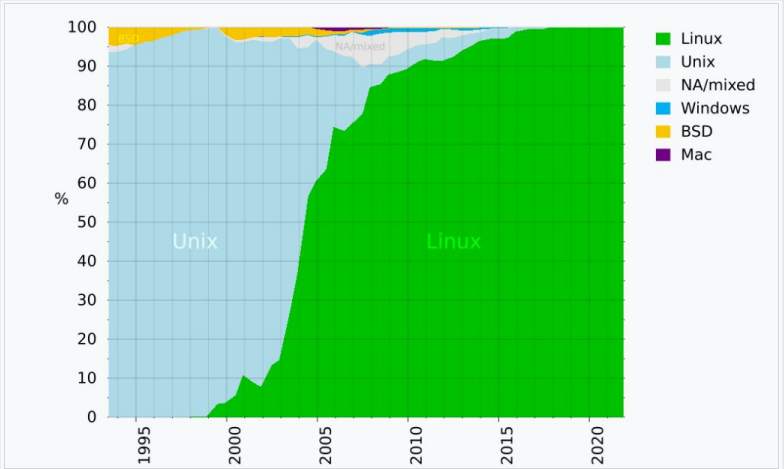

Supercomputer OS family – 1993–2021 systems share according to TOP500[264]

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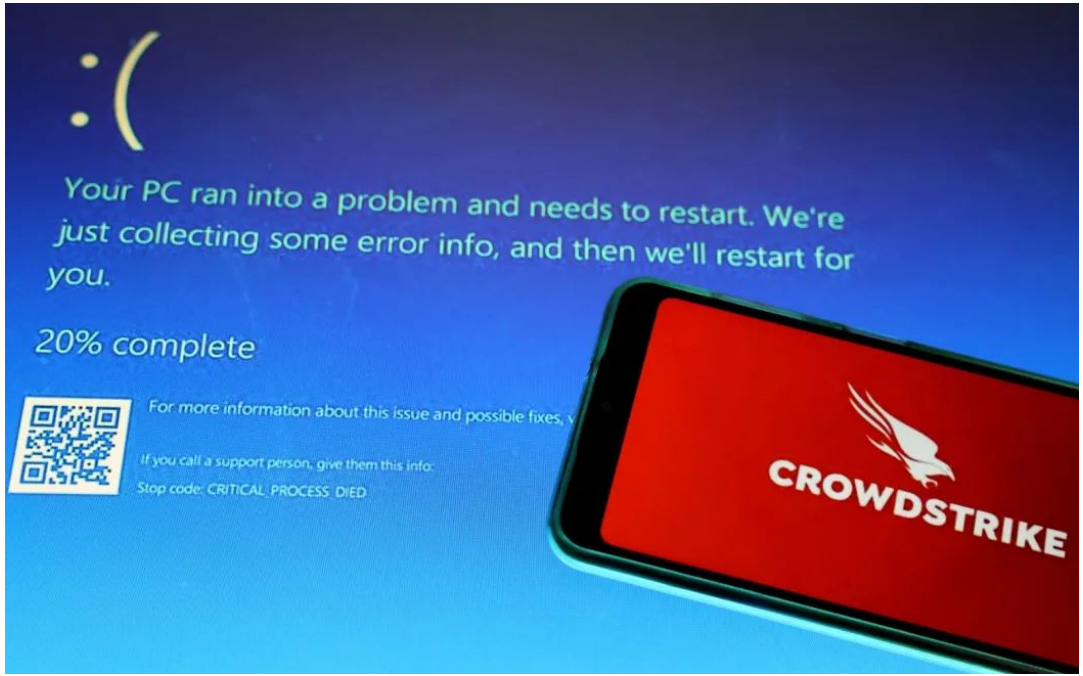

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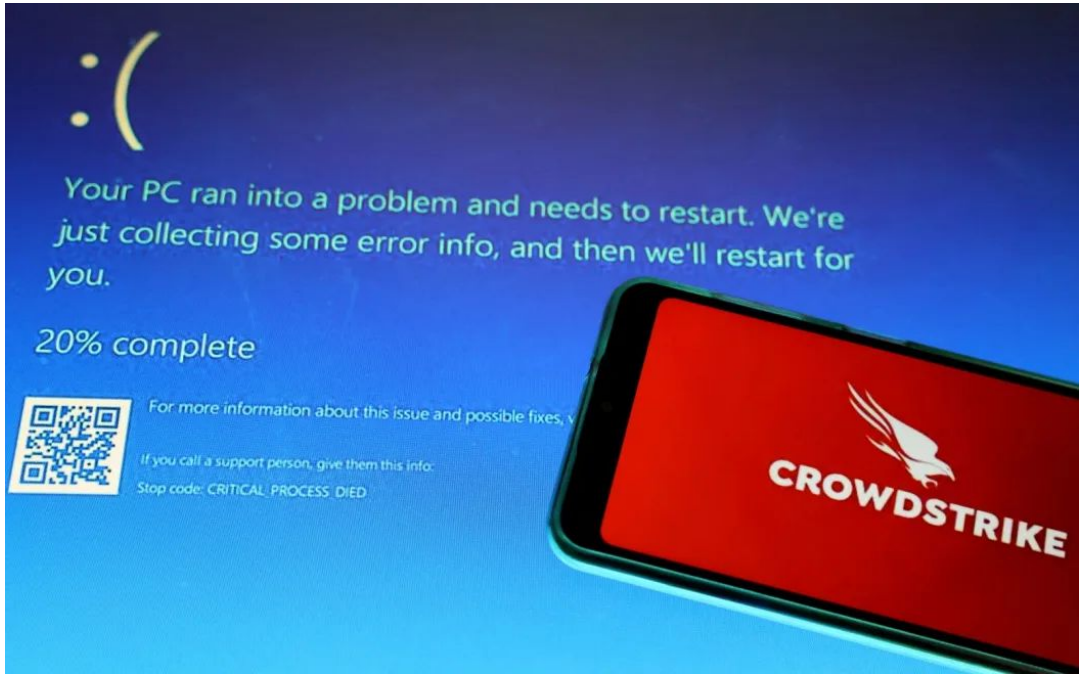
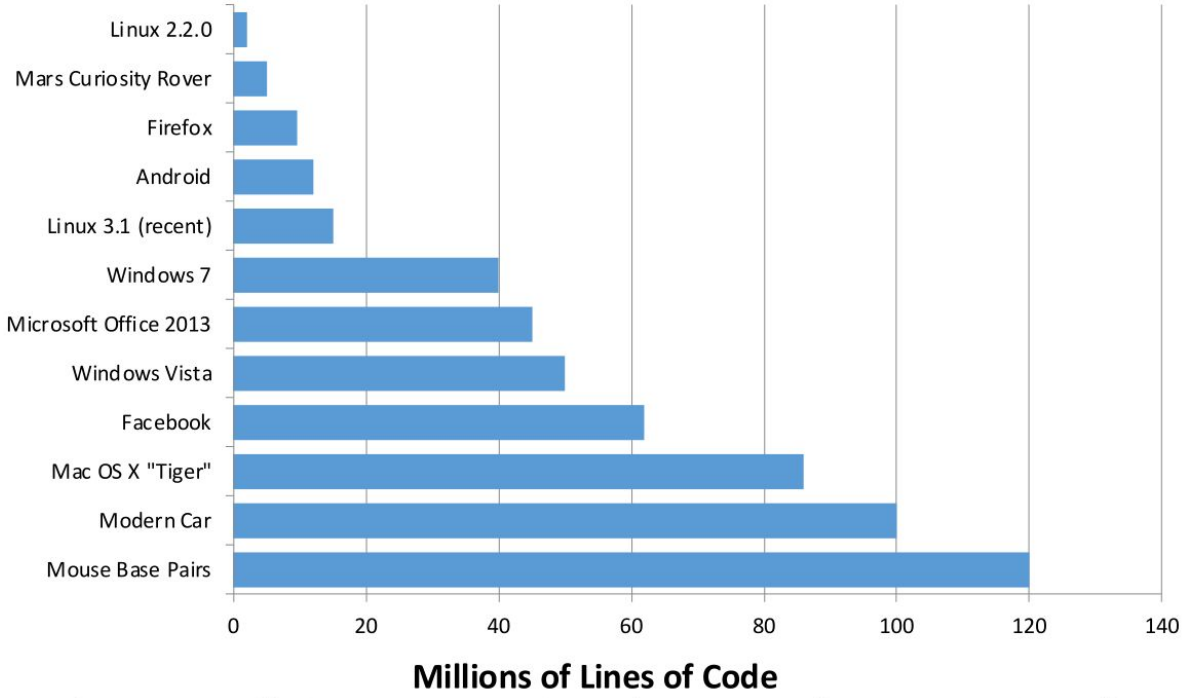


- Reliability
- Efficiency
- Security
- User interface
- Compatibility
- Support

OS complexity



(source https://informationisbeautiful.net/visualizations/million-lines-of-code/)

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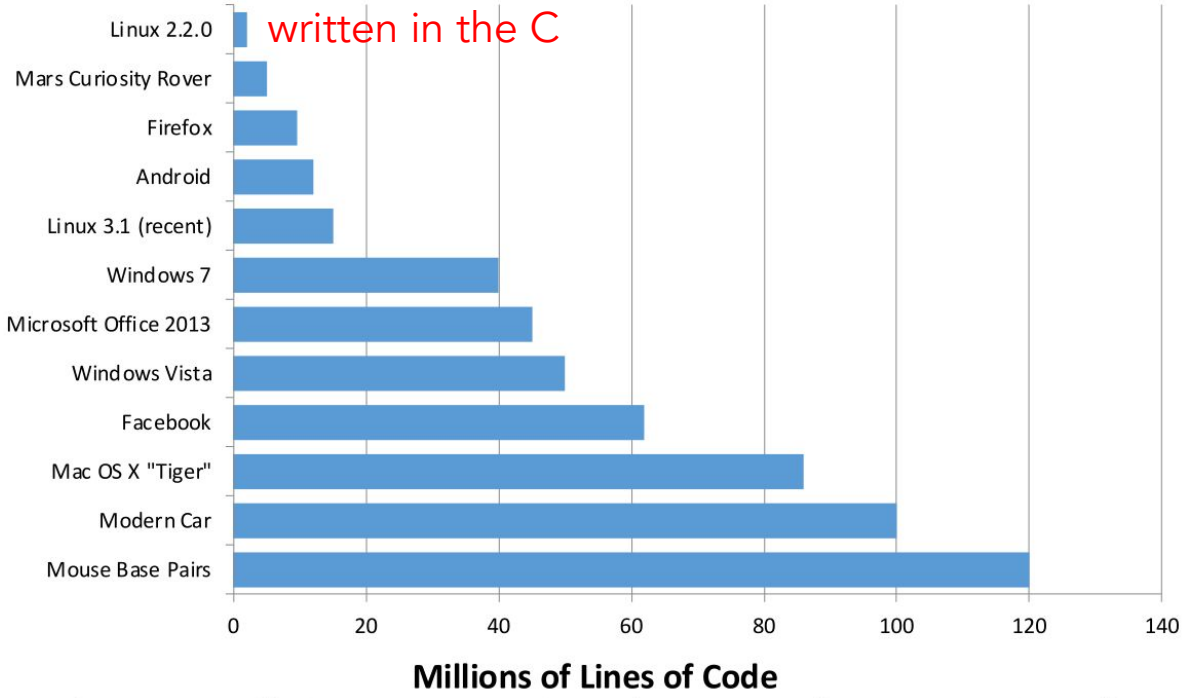

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  - Yes (e.g. build your own Web framework)

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  - Yes (e.g. build your own Web framework)
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  - Hopefully
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Course information

- Lectures
  - Time: 1:15pm - 4:15pm Friday
  - Venue: CR7
- Materials (optional)
  - Operating System Concepts (10th Edition) by Abraham Silberschatz, Peter B. Galvin, and Greg Gagne.
  - Operating Systems: Principles and Practice (2nd Edition) by Thomas Anderson and Michael Dahlin.
  - Operating Systems: Three Easy Pieces by Remzi H Arpaci-Dusseau and Andrea C Arpaci-Dusseau
- Lectures/Homework: Canvas
- Instructor
  - Linh Huynh (linh.huynh@fulbright.edu.vn)
  - Office hours:
    - 15 minutes after the lectures
    - Wednesday (1pm-3pm, appointments by email)

Grading

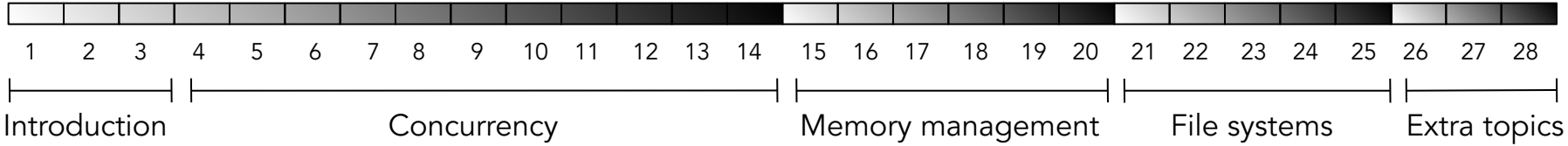
- Homework (6-8 sets) = 35%
  - No late homework
  - Effort-based grading
- Two mid-term exams (45-minute writing) = 30%
- Final exam (75-minute writing) = 25%
- Project = 10%

  100%
- Final letter grades
  - No "curving"
    - A/A- : Can be TAs
    - B+/B/B- : Ready to take advanced courses (e.g. Security, Distributed Systems, etc)
    - C+/C/C- : Not ready to take advanced courses
    - F : Do not understand key concepts & techniques

Tentative schedule

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28

Tentative schedule



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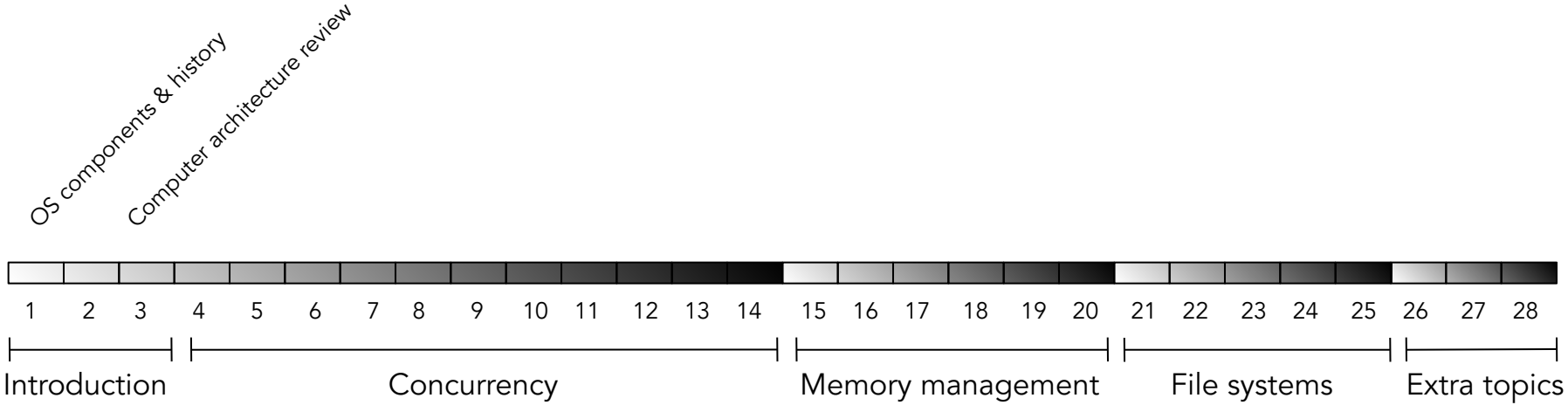

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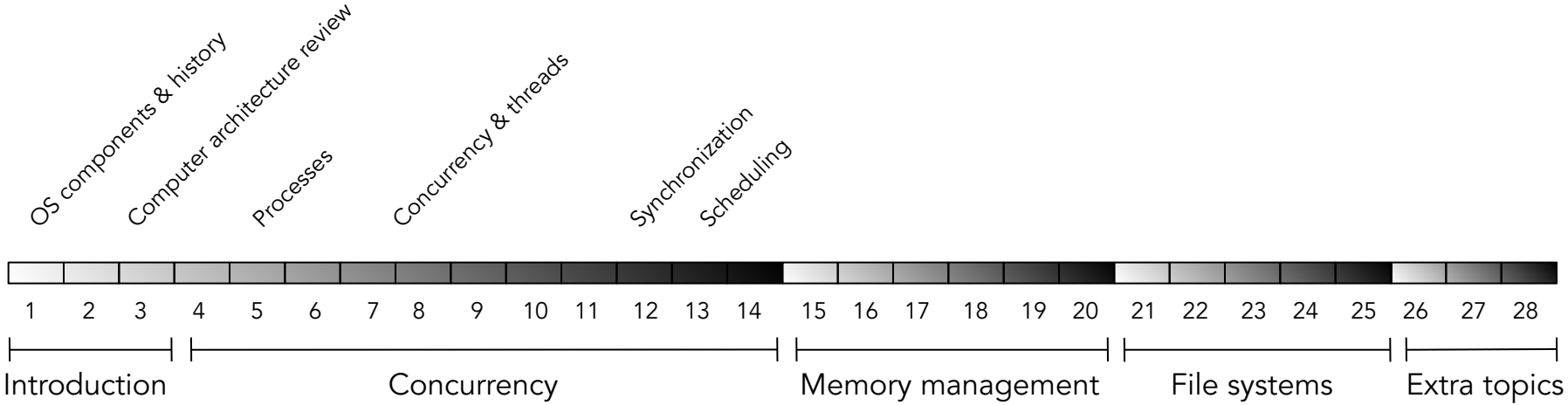

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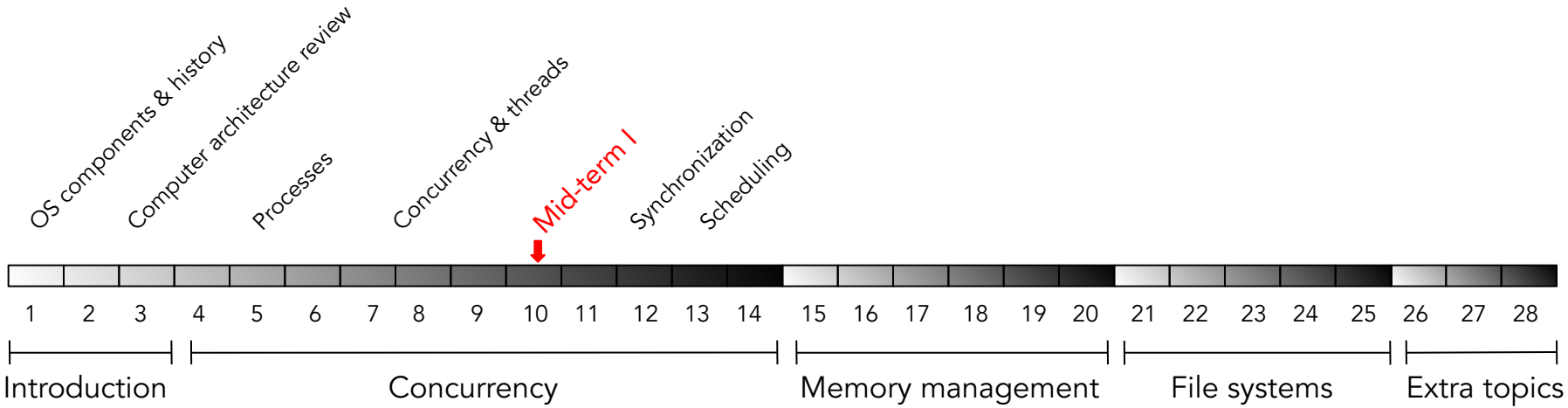

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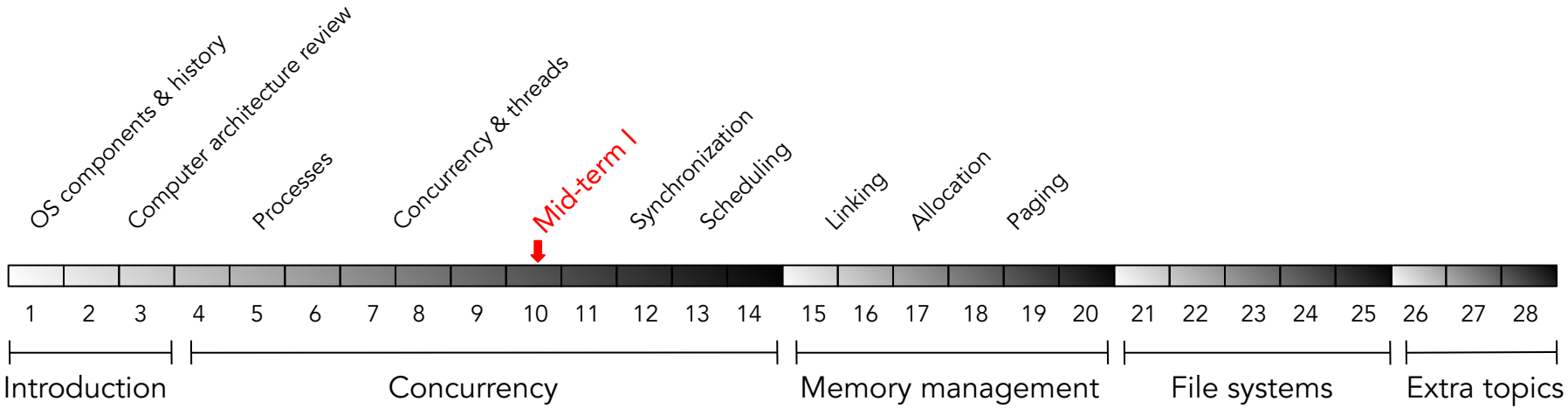

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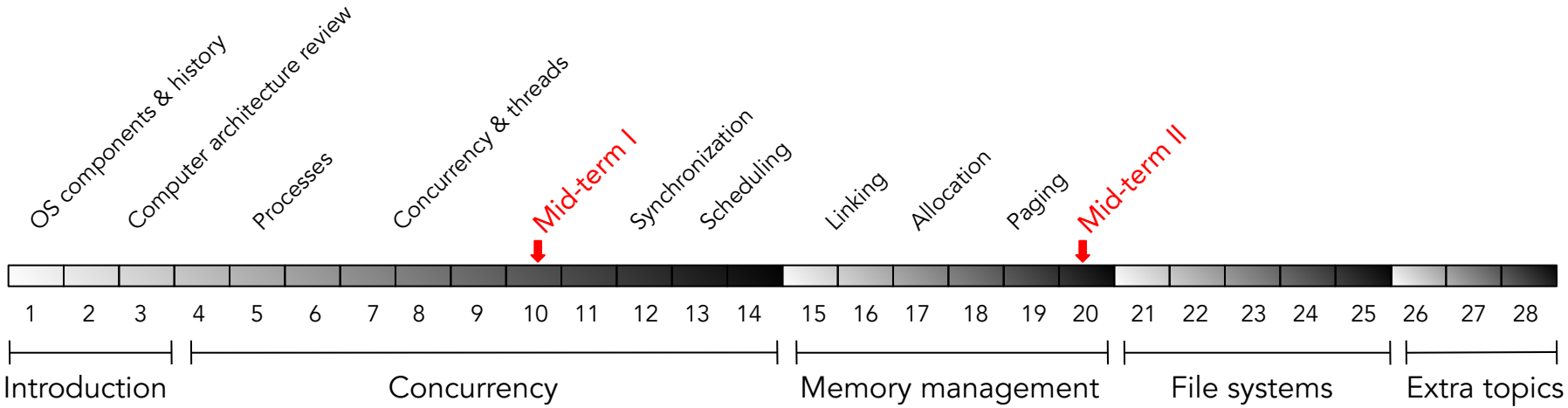

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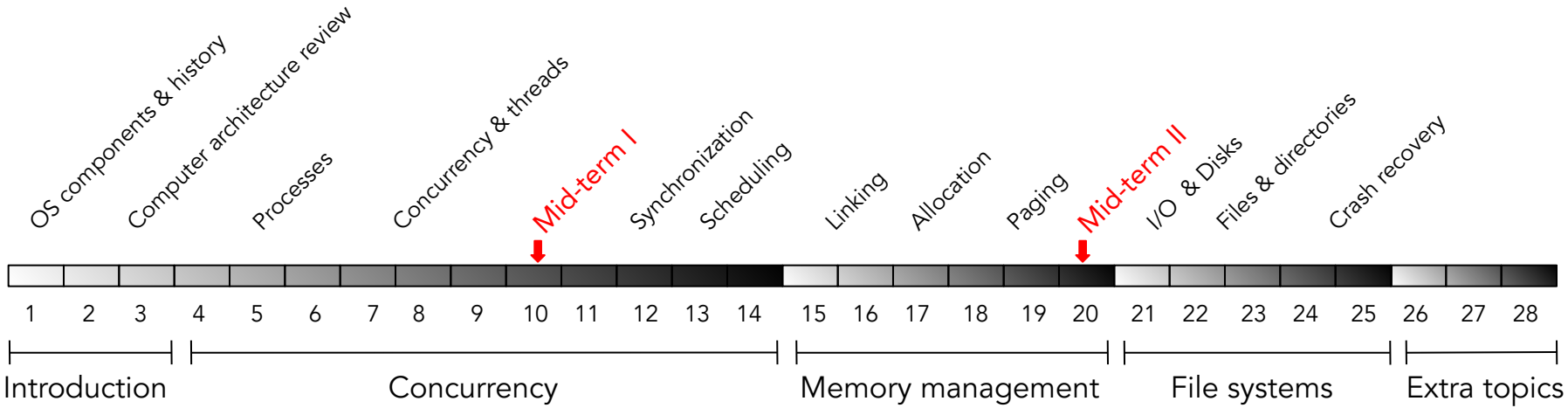

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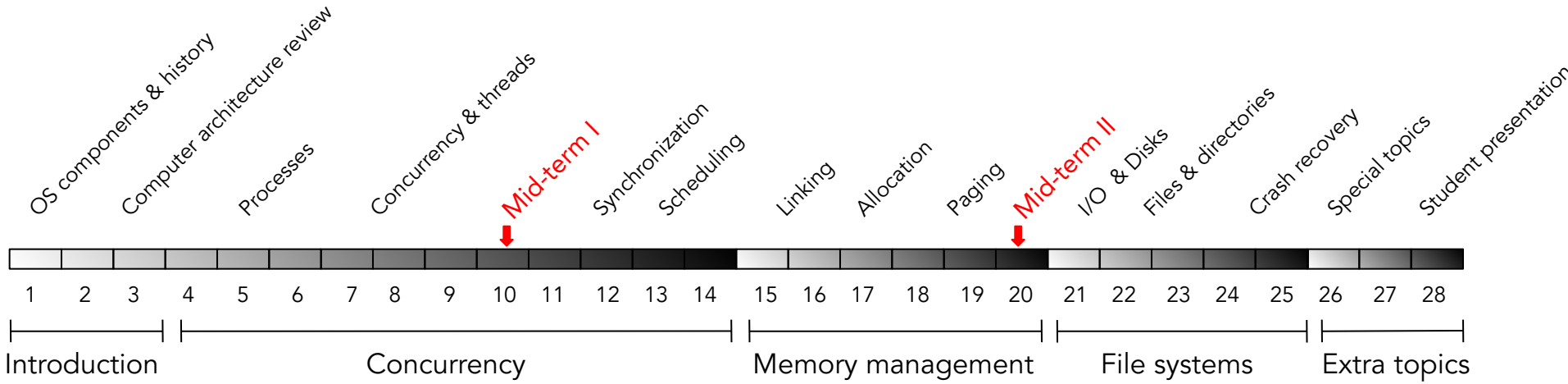

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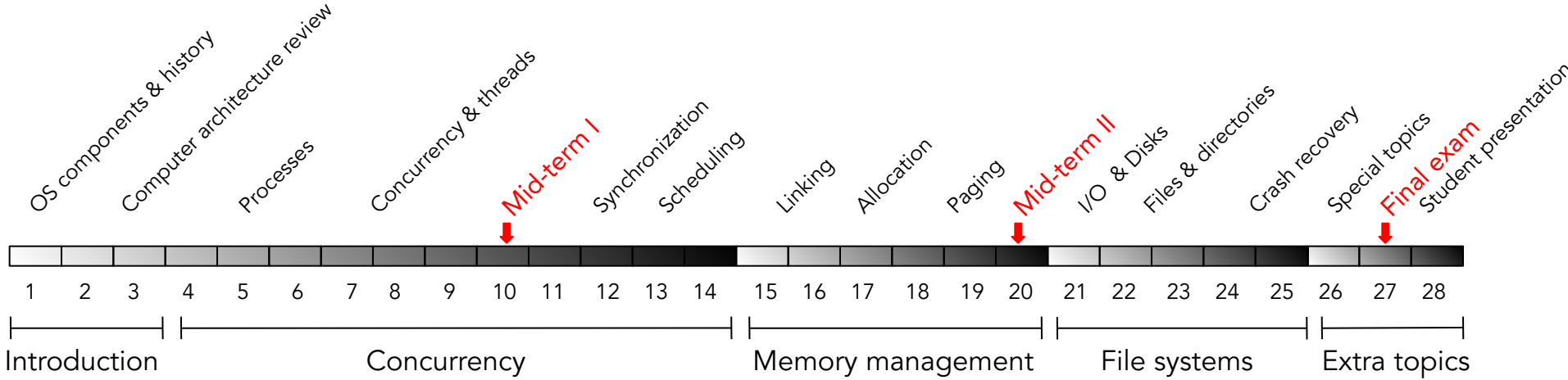

How to study CS211?

- Attend the lecture
  - Ask any questions if you feel confused!
- Do the homework
  - Effort-based grading → Try to solve all problems by yourself
- Connect what you learn in CS211 with
  - Other courses
  - Other projects you have worked or are working on
  - Other information you know about computer systems and softwares
- Ask for the support
  - Urgent questions → Send emails to the instructor
  - Short questions → Ask right after the lectures
  - Long questions → Discuss in office hours

What do you know about Operating Systems & CS211?

1. When did you first hear the term “operating system”?
2. What is an OS?
3. Why do we need OSes?
4. What was the first OS?
5. How many OSes are being used actively?
6. Which OSes do you know or have used?
7. Which OS is the most popular now?
8. Which OS is the most commonly used on servers?
9. What is your favorite OS?
10. Is Windows the favorite OS of the CS211 instructor?
11. What makes a good OS?
12. Are people still developing new OSes?
13. Which programming languages are people using to develop new OSes?
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Summary

- CS211 is
  - not about
    - Advanced functions of Windows?
    - Or how to install and use Linux?
  - but is about
    - What are functions of an OS?
    - What are components of an OS?
    - How are OS components designed and implemented?
    - What are pros and cons of different ways to implement OS components?